



nextwork.org

Continuous Integration with CodeBuild



Amirul Zaol-kefli

```
1  version: 0.2
2
3  phases:
4    install:
5      runtime-version:
6      java: corvetta
7    pre_build:
8      commands:
9        - echo Logging in to AWS CodeArtifact...
10         CODEARTIFACT_AUTH_TOKEN=$(aws codeartifact get-authorization-token --domain nextwork --domain-owner <your_aws_account_id> --region us-east-2 --query authorizationToken --output text)
11         export CODEARTIFACT_AUTH_TOKEN
12    build:
13      commands:
14        - echo Build started on `date`
15        - mvn clean install -s settings.xml
16    post_build:
17      commands:
18        - echo Build completed on `date`
19        - echo Packaging artifacts...
20        - mvn package -s settings.xml
21  artifacts:
22    files:
23      - target/*.war
24    discard-paths: no
25
```



Introducing Today's Project!

In this project, I will demonstrate the way to set up AWS CodeBuild. I'm doing this project to learn how to create continuous integration automation for web apps.

Key tools and concepts

Services I used were CodeArtifact and CodeBuild. Key concepts I learnt include building artifacts using CodeBuild and the actual concept of Continuous Integration.

Project reflection

This project took me approximately 2 hours. The most challenging part was the build process using `buildspec.yml` as my folder structure is different from the tutorial. It was most rewarding to see the build succeed with the changes that I made.

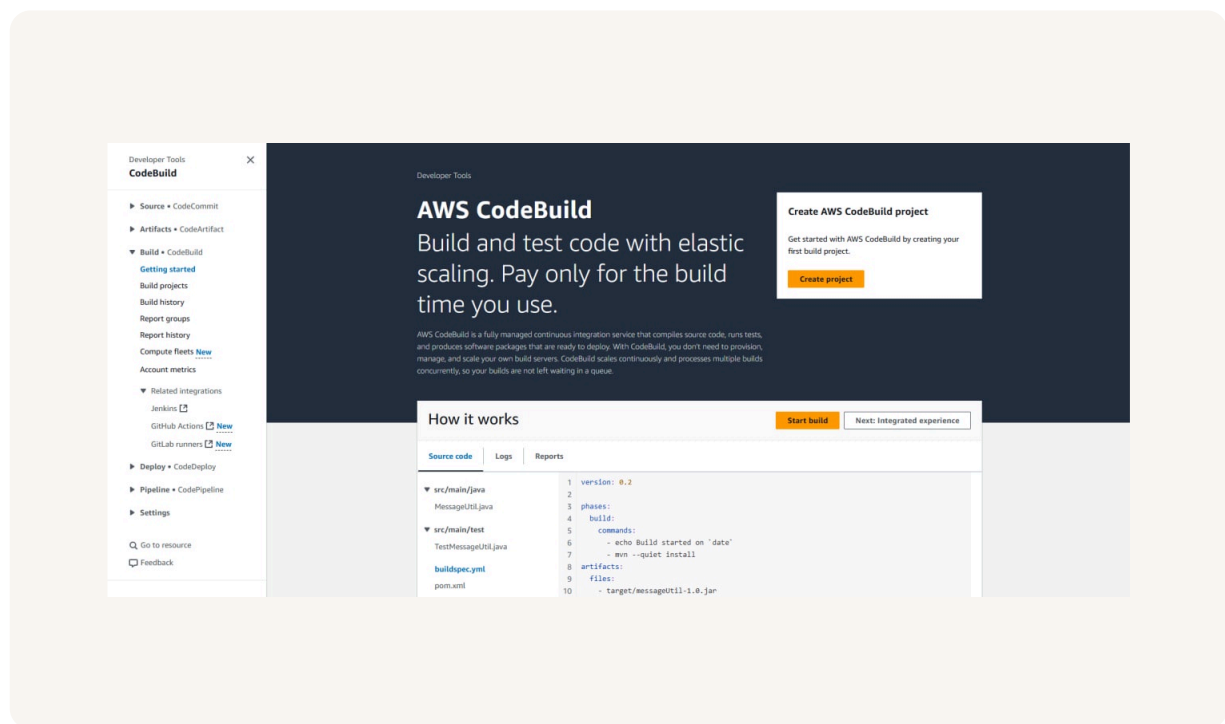
This project is part four of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project next Thursday, where I would attempt Day 5 of the challenge.



Setting up a CodeBuild Project

CodeBuild is a continuous integration service, which means taking my source code, compile it, run tests, and package it up. Engineering teams use it because they don't have to pay for the compute time used for building projects.

My CodeBuild project's source configuration means the location of the code that CodeBuild will fetch, compile, and package into a file I can deploy and I selected Github because that is where I am storing my web app files in.





Connecting CodeBuild with GitHub

There are multiple credential types for GitHub, like Personal Access Token and OAuth App. I used GitHub App because it is generally the simplest and most secure option.

The service that helped connect my AWS account to my Github account is AWS CodeConnections.

The screenshot shows the 'Source' configuration page in the AWS CodeBuild console. At the top, there is a 'Source' section with a dropdown arrow and an 'Add source' button. Below this, 'Source 1 - Primary' is selected. The 'Source provider' is set to 'GitHub'. Under 'Credential', a green checkmark indicates the account is successfully connected by using an AWS managed GitHub App, with a link to 'Manage account credentials'. There is an unchecked checkbox for 'Use override credentials for this project only'. The 'Repository' section has three radio buttons: 'Repository in my GitHub account' (selected), 'Public repository', and 'GitHub scoped webhook'. Below the radio buttons, the repository URL 'https://github.com/iPhantomikX/7DayDevOpsChallenge' is entered in a text box, with a search icon on the left, a close icon on the right, and a refresh icon. The 'Source version' section is optional, with a link to 'optional info' and a text box containing 'main'. At the bottom, there is an 'Additional configuration' section with a link to 'Git clone depth, Git submodules, Build status config'.



CodeBuild Configurations

Environment

My CodeBuild project's Environment configuration means we're using a template that AWS has already created for us. It includes settings like provisioning model, which determines how AWS will set up and manage everything needed for my build.

Artifacts

Build artifacts are tangible outputs of my build process. They're important because they are what I'll actually deploy to my servers or distribute to users. My build process will create build artifacts. To store them, I created an S3 bucket.

Packaging

When setting up CodeBuild, I also chose to package artifacts in a zip because it is smaller, organized and simple.

Monitoring

For monitoring, I enabled CloudWatch Logs, which is a monitoring service that collects and tracks logs from AWS services, incredibly useful for debugging and understanding what went wrong if a build fails.



buildspec.yml

My first build failed because it could not locate a buildspec.yml file. A buildspec.yml file is needed because it is a step-by-step instruction manual that runs phases in order - install, pre_build, build, then post_build.

The first two phases in my buildspec.yml file installs java and initialize environment. The third phase in my buildspec.yml file compiles and builds the code. The fourth phase in my buildspec.yml file packages the build.

```
1 version: 0.2
2
3 phases:
4   install:
5     runtime-versions:
6       java: corretto8
7   pre_build:
8     commands:
9       - echo logging in to AWS CodeArtifact...
10      - CODEARTIFACT_AUTH_TOKEN=$(aws codeartifact get-authorization-token --domain nextwork --domain-owner <your_aws_account_id> --region us-east-2 --query authorizationToken --output text)
11      - export CODEARTIFACT_AUTH_TOKEN
12   build:
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16   post_build:
17     commands:
18       - echo Build completed on `date`
19       - echo Packaging artifacts...
20       - mvn package -s settings.xml
21 artifacts:
22   files:
23     - target/*.war
24   discard-paths: no
25
```



Success!

My second build also failed, but with a different error that said `UPLOAD_ARTIFACTS` failed due to no matching artifact paths found in Phase details. To fix this we need CodeBuild to have permission to access CodeArtifact.

To resolve the second error, I had to provide permission access for CodeBuild to access CodeArtifact. When I built my project again, I saw it has successfully downloaded and build the code.

To verify the build, I checked the S3 bucket. Seeing the artifact tells me that the CodeBuild has successfully build and packaged the CodeArtifact into a zip file which contains the `nextwork-web-project.war` file.



Build status			
Status ✔ Succeeded	Initiator amirul-iam-admin	Build ARN arn:aws:codebuild:ap-southeast-1:968626156119:build/nextwork-devops-cicd:dd7335a3-e984-42f5-97c5-78042f03247a	Resolved source version fd915a7619db5ffb5b4ede7ea92b8d9595f089c6
Start time Oct 16, 2025 4:10 PM (UTC+8:00)	End time Oct 16, 2025 4:11 PM (UTC+8:00)	Build number 10	



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